

Grow Expense App

Complete Technical Architecture, Feature Blueprint & User Manual

Project Name:	Grow Expense (Personal & Business Financial OS)	Version:	3.5.0 (Release Build)
Framework:	Flutter 3.x / Dart 3.x	Local DB:	SQLite (sqflite) Offline-First
Cloud Backend:	Supabase PostgreSQL & Realtime Auth	AI Engine:	Google Gemini 2.5 Flash & NVIDIA LLaMA 3.3
Multi-Lingual:	8 Indian Languages + Hinglish Support	Platforms:	Android (Universal & Split ABIs), iOS, Web

Executive Overview:

Grow Expense is a state-of-the-art financial management ecosystem designed specifically for individuals, freelancers, retail shopkeepers, and micro/small business owners. Unlike standard single-purpose expense trackers, Grow Expense is engineered with a **Dual-Engine Architecture** that allows users to toggle instantly between **Personal Mode** (expense logging, visual budget analytics, recurring subscriptions, SMS parsers, predictive forecasting) and **Business Mode** (GST/Non-GST invoicing, inventory management with barcode/QR generation, customer/supplier Khata ledger, automated payment reminders via WhatsApp, and daily P&L; statements).

Built with a robust **Offline-First approach**, data is instantaneously accessible and persisted locally in SQLite, and asynchronously synchronized with Supabase Cloud with automatic conflict resolution, zero data loss on logout/login, and military-grade biometric protection.

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1. System Architecture & Core Technology Stack

Grow Expense is developed using a multi-tiered reactive architecture ensuring sub-16ms UI responsiveness (60-120 FPS high refresh rate support), minimal battery footprint, and zero latency local persistence.

Layer	Technology / Framework	Key Responsibilities & Capabilities
Frontend UI	Flutter 3.x / Dart 3.x	Material Design 3 with custom glassmorphism, dynamic dark/light themes, Google Fonts (Outfit & Inter), responsive layouts for phones & tablets.
State Management	Provider & MultiProvider	Granular reactive state isolation across UserProvider, ExpenseProvider, BusinessProvider, and ThemeController.
Local Persistence	SQLite (sqflite) + SharedPreferences	Zero-latency offline read/write. Stores transactions, khata entries, invoices, inventory items, and cached AI chat histories.
Cloud Backend	Supabase (PostgreSQL 15)	Realtime Auth, PostgreSQL Row-Level Security (RLS), incremental timestamp sync, soft-delete propagation, and automated cloud backup.
AI Engines	Google Gemini 2.5 Flash / NVIDIA LLaMA 3.3	Context-aware financial analysis, autonomous ledger actions (add expense, split bills, set budgets), NLP prompt parsing.
Hardware & Voice	Speech-To-Text (STT) + Local Biometrics	Multi-lingual voice expense recording, Sabji Mandi voice rate interpreter, Fingerprint / Face Unlock security.

2. Dual-Engine Operating System (Personal vs. Business)

A cornerstone feature of Grow Expense is its seamless one-tap switching between **Personal Mode** and **Business Mode**. The entire application dynamically restructures its dashboard, navigation bars, reporting metrics, and AI prompts depending on the active mode.

Feature / Aspect	Personal Mode ■■	Business Mode ■■■
Primary Purpose	Personal budgeting, expense tracking, daily spend limits, savings growth.	Shopkeeper POS, daily sales turnover, invoice generation, customer credits (Udhar).
Dashboard Metrics	Total Spent, Monthly Budget Remaining, Category Breakdown, Recurring Bills.	Total Revenue/Sales, Cost of Goods, Net Profit/Loss, Total To Receive & To Pay.
Key Modules	Budget Goals, Voice Expense Entry, SMS Parser, Mandi Unit Calculator, Analytics.	Tax & Quotation Invoices, Khata Book, Product Catalog, Barcode Generator, POS.
AI CFO Advisor	Personalized saving tips, 50/30/20 budget analysis, overspending alerts.	Business profit margins, tax liabilities, slow-moving inventory alerts, credit risk.
Customer/Party CRM	Friend debt splitting & informal loans.	Full Customer & Supplier Ledger with WhatsApp payment reminder links & PDF bills.

3. Personal Finance & Wealth Modules

1. Smart Multi-Category Expense & Income Tracker:

- Instant logging with predefined & customizable categories (Food, Travel, Bills, Shopping, Health, Salary, Investments). Supports multi-payment modes (UPI, Cash, Credit Card, Net Banking).
- Visual receipt and invoice photo attachments with full-screen zoom and cloud persistence.
- Tagging system (#vacation, #party, #medical) for cross-category granular audits.

2. Automated SMS Financial Parser (Offline NLP):

- Intelligent regex engine parses bank transaction SMS messages (HDFC, SBI, ICICI, Axis, Paytm, GPay, PhonePe) securely on-device without sharing SMS contents to external servers.
- One-tap approval to convert detected debits and credits into verified ledger entries.

3. Visual Budgeting & Spending Prediction:

- Category-specific and monthly aggregate budget caps with real-time progress bars (Green < 70%, Amber 70-90%, Red > 90%).
- Machine learning linear regression prediction estimates end-of-month spending based on current burn rate.

4. Recurring Subscriptions & Utility Reminders:

- Automatic scheduling for rent, electricity, Netflix, SIPs, and insurance premiums with notification alarms.

4. Business Management, Invoicing & Khata Ledger

1. Professional GST & Non-GST Invoicing Engine:

- Create, customize, and print beautiful PDF invoices, estimates/quotations, and delivery challans.
- Configurable GST rates (0%, 5%, 12%, 18%, 28%), IGST/CGST/SGST splitting, and HSN/SAC codes.
- Include business logo, signature, bank UPI QR code for instant customer payment, and terms & conditions.
- Direct 1-tap WhatsApp sharing with formatted PDF documents and auto-generated message text.

2. Digital Khata Book (Customer & Supplier Credit Ledger):

- Complete replacement for paper udhar books. Track 'Maine Diye' (Gave/Credit) and 'Maine Liye' (Got/Debit).
- Free WhatsApp payment reminder integration with customizable regional language greetings and UPI payment links.
- Comprehensive party statement PDF generation with date-range filters and running balance calculations.

3. Inventory & Barcode/QR Label Generator:

- Product catalog management with SKU, cost price, selling price, low stock thresholds, and unit types.
- Built-in printable barcode and QR label generator with custom sheet layouts (A4 24-up, 40-up, thermal rolls).
- Camera-based barcode scanner for rapid point-of-sale checkout and stock deduction.

5. Mandi & Financial Calculator Hub

Grow Expense includes a specialized 5-in-1 financial calculator suite designed for Indian markets, vegetable/grocery mandi traders, and everyday financial calculations:

Calculator Tool	Description & Capabilities	Voice & Regional Support
Sabji Mandi Unit Rate Calculator	Computes exact rates for awkward weights (e.g., 250g, 500g, 1.5kg, 3.75kg based on ■/kg or ■/quintal). Instant budget-to-weight reverse calculation.	AI Voice input in Hindi, Bengali, Marathi, Gujarati, Tamil, Telugu, Kannada & English.
Loan & EMI Calculator	Calculates monthly EMI, total interest, and full repayment amortization schedule for Home, Car, and Personal Loans.	Interactive sliders + direct tap-to-type numerical input with instant breakdown pie chart.
GST Tax Calculator	1-tap Add GST (+) or Remove GST (-) for all standard Indian tax slabs (3%, 5%, 12%, 18%, 28%).	Shows original amount, GST component, and net total with quick-copy clipboard button.
Discount & Sale Calculator	Calculates discounted price, percentage saved, and multi-tier promotional discounts.	Instant calculation with history logging.
SIP & Wealth Compounder	Projects mutual fund SIP growth, total invested amount, and estimated future wealth.	Visual wealth vs invested capital comparison chart.

6. Autonomous AI CFO & Intelligent Financial Advisor

Powered by Google Gemini 2.5 and NVIDIA LLaMA 3.3, **Grow Expense AI** acts as a personalized Virtual Chief Financial Officer with real-time database action capabilities.

- Key AI Capabilities & Autonomous Action Engine:**
- **Live Financial Ledger Awareness:** Automatically injects real-time financial snapshots (current month's spend, top categories, profit margins, outstanding receivables) into the prompt context without exposing PII.
 - **Autonomous Action Execution:** When the user types or speaks natural language commands (e.g. *'Add ₹500 for petrol'*, *'Set food budget to ₹6,000'*, *'Split ₹1,200 dinner with Rahul'*), the AI generates an interactive Action Proposal Card that users can execute with 1 tap directly into SQLite.
 - **Multi-Session Chat Architecture:** ChatGPT-style session drawer allowing users to create new topics, rename sessions, search chat histories, and clear cache.
 - **Dual Model Redundancy & Hybrid Key System:** Users can choose zero-config server default AI or supply their own custom Gemini/NVIDIA API keys with seamless automated fallback if one service experiences rate limits.

7. Multi-Lingual Regional Support Matrix (8 Languages)

Grow Expense provides native localized UI strings, voice recognition locales, notification templates, and AI prompt formatting across 8 major Indian languages:

Language	Code	Native Script	Voice STT Locale	AI & Calculator Synchronization
English	en	English	en_IN	Full UI, Prompts, Voice, Notifications
Hindi	hi	हिन्दी	hi_IN	Full UI, Prompts, Voice, Notifications
Bengali	bn	বাংলা	bn_IN	Full UI, Prompts, Voice, Notifications
Marathi	mr	मराठी	mr_IN	Full UI, Prompts, Voice, Notifications
Gujarati	gu	ગુજરાતી	gu_IN	Full UI, Prompts, Voice, Notifications
Tamil	ta	தமிழ்	ta_IN	Full UI, Prompts, Voice, Notifications
Telugu	te	తెలుగు	te_IN	Full UI, Prompts, Voice, Notifications
Kannada	kn	ಕನ್ನಡ	kn_IN	Full UI, Prompts, Voice, Notifications
Hinglish	hinglish	Colloquial Mix	en_IN / hi_IN	Conversational Romanized Hindi Prompts

8. Offline-First Synchronization & Conflict Resolution

Grow Expense utilizes an advanced bidirectional synchronization engine ensuring 100% offline availability with instant cloud consistency upon internet reconnection:

1. Incremental Timestamp Sync & Soft Deletes:

- Every table contains `updated\_at` (ISO8601 UTC) and `is\_deleted` (0 or 1).
- Deletions are soft-deleted locally and pushed to Supabase to guarantee cross-device consistency without leaving orphan records.

2. Zero Data Loss Authentication Lifecycle:

- **Account Logout:** Safely triggers an immediate push of any pending offline records, followed by a clean wipe of local SQLite caches to prevent account data leakage between multiple users on the same device.
- **Fresh Login / Reinstall:** Automatically fetches the user's complete cloud ledger (expenses, budgets, khata entries, business sales, invoices) and populates SQLite seamlessly within seconds.

3. Conflict Resolution Strategy:

- Last-Write-Wins (LWW) based on server-side PostgreSQL timestamp verification.
- UUIDv4 primary keys prevent ID collisions across offline client creations.

9. Security, Biometrics, Export & Cloud Backup Lifecycle

1. Enterprise Security & Privacy:

- Biometric authentication (Fingerprint / Face ID) with auto-lock timer.
- Supabase Row-Level Security (RLS) ensures users can only read/write their own authenticated records.
- API Keys stored strictly in encrypted SharedPreferences on-device.

2. Export & Reporting:

- **PDF Financial Statements:** Clean, formatted summary with category distribution charts and transaction line items.
- **Excel / CSV Export:** Raw ledger export for tax auditors and Chartered Accountants.
- **Full JSON Backup & Restore:** Offline scope-selectable encrypted backup file.

10. Database Schema Specifications & Data Models

Below is the core relational schema implemented across local SQLite and Supabase PostgreSQL:

Table Name	Primary Columns	Description & Relationships
expenses	id (UUID), title, amount, category, date, payment_mode, notes, is_recurring, user_id, is_deleted, updated_at	Stores personal expenses with receipt links and recurring subscription flags.
budgets	id (UUID), category, amount_limit, month_year, alert_threshold, user_id, is_deleted, updated_at	Monthly budget caps per category for personal financial guardrails.
business_sales	id (UUID), invoice_no, customer_name, customer_phone, total_amount, tax_amount, discount, payment_status, date, user_id	Business POS and invoicing transactions with tax & payment statuses.
khata_entries	id (UUID), party_name, party_phone, party_type (customer/supplier), entry_type (you_gave/you_got), amount, date, user_id	Digital Udhar ledger recording receivables and payables per contact.
inventory_products	id (UUID), name, sku_barcode, purchase_price, selling_price, stock_quantity, unit_type, min_alert_qty, user_id	Product catalog and stock management with barcode scanner integration.
ai_chat_messages	id (UUID), session_id, message_text, is_user, timestamp, action_intent_json, mode (personal/business)	Multi-session AI chat conversation history with action intent persistence.

11. Setup, Environment Configuration & Build Instructions

1. Prerequisites:

- Flutter SDK: 3.22.x or higher (Dart 3.4.x+)
- Android Studio / VS Code with Flutter extensions
- JDK 17 (Java Development Kit)
- Android SDK Platform 34 (API Level 34)

2. Environment & Secret Setup:

Configure Supabase credentials in frontend/lib/services/supabase_service.dart:

```
SUPABASE_URL = 'https://[your-project-id].supabase.co'
```

```
SUPABASE_ANON_KEY = 'eyJhbGciOiJIUzI1Ni...'
```

3. Compiling Release Production APKs:

To generate high-efficiency split APKs per architecture (ARM64, ARMv7, x86_64):

```
flutter clean && flutter pub get
```

```
flutter build apk --release --split-per-abi
```

Output Artifacts:

- ARM64-v8a (Modern 64-bit phones):** build/app/outputs/flutter-apk/app-arm64-v8a-release.apk
- ARMeabi-v7a (Older 32-bit phones):** build/app/outputs/flutter-apk/app-armeabi-v7a-release.apk
- x86_64 (Emulators / ChromeOS):** build/app/outputs/flutter-apk/app-x86_64-release.apk